
Addition and Subtraction of Rational Expressions with the Same Denominator

Note: To add or subtract rational expressions, the expressions must have the same denominator.

PROPERTY Addition and Subtraction of Rational Expressions

Let p , q , and r represent polynomials where $q \neq 0$. Then,

$$1. \quad \frac{p}{q} + \frac{r}{q} = \frac{p+r}{q}$$

$$2. \quad \frac{p}{q} - \frac{r}{q} = \frac{p-r}{q}$$

Note: To add or subtract rational expressions with the same denominator:

- combine the terms in the numerator
- write the result over the common denominator
- if possible, simplify the expression to lowest terms

For exercises 1 – 6, add or subtract the expressions with like denominators as indicated.

$$1. \quad \frac{7}{10} - \frac{2}{10}$$

$$2. \quad \frac{3a}{a-4} - \frac{a+8}{a-4}$$

$$3. \quad \frac{4c}{c+5} + \frac{20}{c+5}$$

$$4. \quad \frac{d^2}{d-1} - \frac{8d-7}{d-1}$$

$$5. \quad \frac{c^2}{c-6} - \frac{36}{c-6}$$

$$6. \quad \frac{4}{3x^2 + 2x - 8} + \frac{-3x}{3x^2 + 2x - 8}$$

Addition and Subtraction of Rational Expressions with Different Denominators

Note: To add or subtract two rational expressions, the expressions must have the same denominator.

PROCEDURE Adding or Subtracting Rational Expressions

Step 1. Factor the denominators of each rational expression.

Step 2. Identify the LCD.

Step 3. Rewrite each rational expression as an equivalent expression with the LCD as its denominator.

Step 4. Add or subtract the numerators, and write the result over the common denominator.

Step 5. Simplify.

For exercises 7 – 13, add or subtract the expressions with unlike denominators as indicated.

$$7. \quad \frac{4}{a^2b^4} + \frac{2}{a^4b^3}$$

$$8. \quad \frac{4}{5t+10} + \frac{6}{t+2}$$

$$9. \quad \frac{y}{y-8} + \frac{4}{y}$$

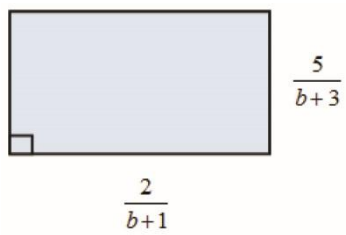
$$10. \quad \frac{24}{m^2-4m} - \frac{3m}{2m-8}$$

$$11. \quad \frac{3}{x^2+5x+6} + \frac{3}{x^2+7x+12}$$

$$12. \quad \frac{p-3}{p^2+3p+2} + \frac{p-1}{p^2-4}$$

13.
$$\frac{2}{c+2} - \frac{3}{c} + \frac{c+10}{c^2-4}$$

14. Find an expression that represents the perimeter of the figure. Assume that $b > 0$.



Using Rational Expressions in Translations

15. Write the reciprocal of the difference of a number and 4.
16. Write the quotient of 6 and the sum of 3 and a number.

For exercises 17 and 18, translate the English phrases into algebraic expressions. Then simplify by combining the rational expressions.

17. The sum of the reciprocal of a number and the quotient of 5 and twice the number.

18. The difference of a number and six times the reciprocal of the number.